

Brac University

Department of Electrical & Electronic Engineering

Semester Spring-25

Course Number: EEE103L

Course Title: Computer Programming Laboratory

Section: 06

Lab Report



Experiment no.

Name of the experiment: Familiarization with Functions and Recursion

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Lab report of experiment 4

a. Write a C program to find whether a number is Palindrome or Not using functions.

Code:

```
#include <stdio.h>

int reverse_num(int num)
{
    int reverse = 0;
    while (num > 0)
    {
        reverse = reverse * 10 + num % 10;
        num = num/10;
    }
    return reverse;
}

int palindrome(int num)
{
    return num == reverse_num(num);
}

int main()
{
    int number;

    printf("Enter a number: ");

    scanf("%d", &number);

    if(palindrome(number))
        printf("%d is a Palindrome number.\n", number);
```

else

```
printf("%d is Not a Palindrome number.\n", number);
```

```
return 0;
```

```
}
```

Screenshot of code:



```
C palindrome.c •
D: > BracU > 4. Fourth Semester (SP-25) > EEE103 > Lab report > exp 4 > C palindrome.c > main()
1  #include <stdio.h>
2
3  int reverse_num(int num)
4  {
5      int reverse = 0;
6      while (num > 0)
7      {
8          reverse = reverse * 10 + num % 10;
9          num = num/10;
10     }
11     return reverse;
12 }
13
14 int palindrome(int num)
15 {
16     return num == reverse_num(num);
17 }
18
19 int main()
20 {
21     int number;
22
23     printf("Enter a number: ");
24     scanf("%d", &number);
25
26     if(palindrome(number))
27         printf("%d is a Palindrome number.\n", number);
28     else
29         printf("%d is Not a Palindrome number.\n", number);
30
31     return 0;
32 }
33
```

Output:

Is a Palindrome number

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL

PS C:\Users\aryan> & 'c:\Users\aryan\.vscode
in=Microsoft-MIEngine-In-hk0vnldn.mbj' '--sto
icrosoft-MIEngine-Pid-0icwsknq.snm' '--dbgExe
Enter a number: 1331
1331 is a Palindrome number.
```

Is not a Palindrome number

```
PS C:\Users\aryan> & 'c:\Users\ary
in=Microsoft-MIEngine-In-cxrijdbby.b
icrosoft-MIEngine-Pid-ynixpfl5.a0n'
Enter a number: 1246
1246 is Not a Palindrome number.
```

b. Write a C program to get the nth Fibonacci Term using recursion.

Code:

```
#include <stdio.h>

int f(int n)
{
    if (n == 0)
        return 0;
    else if (n == 1)
        return 1;
    else
        return f(n - 1) + f(n - 2);
}
```

```

int main()

{

int n;

printf("Enter the (n) position to get the Fibonacci term: ");

scanf("%d", &n);

if (n < 0)

printf("Please enter a positive integer.\n");

else {

int result = f(n);

printf("Fibonacci term at position %d is %d\n", n, result);

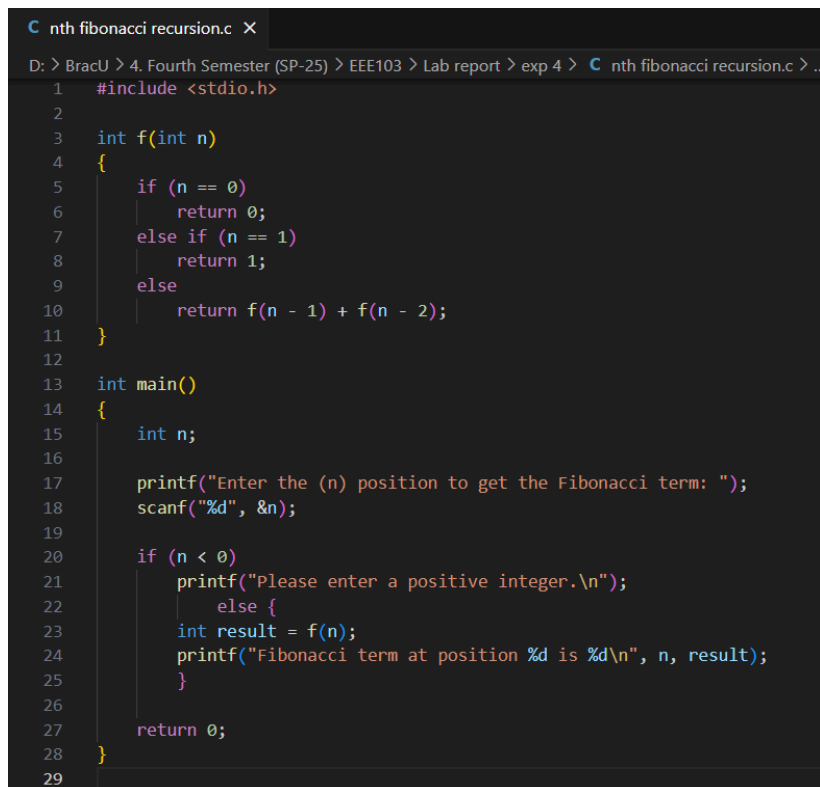
}

return 0;

}

```

Screenshot of code:

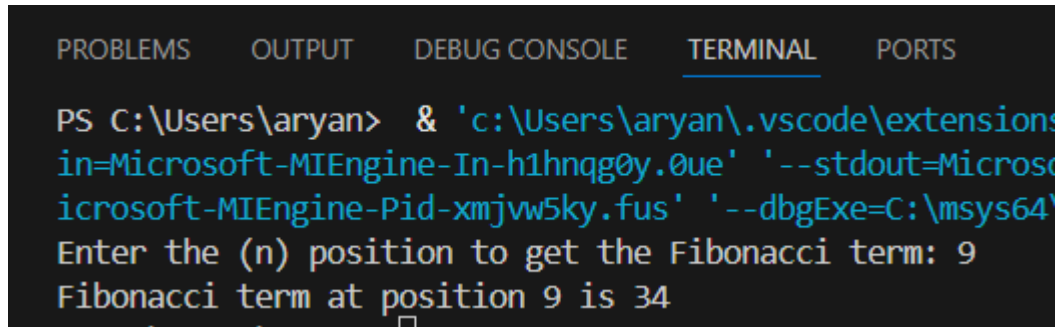


```

C nth fibonacci recursion.c X
D: > BracU > 4. Fourth Semester (SP-25) > EEE103 > Lab report > exp 4 > C nth fibonacci recursion.c > ..
1  #include <stdio.h>
2
3  int f(int n)
4  {
5      if (n == 0)
6          return 0;
7      else if (n == 1)
8          return 1;
9      else
10         return f(n - 1) + f(n - 2);
11 }
12
13 int main()
14 {
15     int n;
16
17     printf("Enter the (n) position to get the Fibonacci term: ");
18     scanf("%d", &n);
19
20     if (n < 0)
21         printf("Please enter a positive integer.\n");
22     else {
23         int result = f(n);
24         printf("Fibonacci term at position %d is %d\n", n, result);
25     }
26
27     return 0;
28 }
29

```

Output:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\aryan> & 'c:\Users\aryan\.vscode\extensions\Microsoft-MIEngine-In-h1hnqg0y.0ue' '--stdout=Microsoft-MIEngine-Pid-xmjvw5ky.fus' '--dbgExe=C:\msys64\
Enter the (n) position to get the Fibonacci term: 9
Fibonacci term at position 9 is 34
```

c. Write a C program to find the sum of digits of a given number using recursion.

Code:

```
int main()
{
    int n;

    printf("Enter a number: ");

    scanf("%d", &n);

    if (n < 0)

        printf("Please enter a positive number.\n");

    else {

        int result = sumdigits(n);

        printf("Sum of digits of %d is %d\n", n, result);

    }

    return 0;
}
```

Screenshot of code:

```
sum of d recursion.c X
D: > BracU > 4. Fourth Semester (SP-25) > EEE103 > Lab report > exp 4 > C sum of d recursion.c > ...
1  #include <stdio.h>
2
3  int sumdigits(int num)
4  {
5      if (num == 0)
6          return 0;
7      else
8          return (num % 10) + sumdigits(num / 10);
9  }
10
11 int main()
12 {
13     int n;
14
15     printf("Enter a number: ");
16     scanf("%d", &n);
17
18     if (n < 0)
19         printf("Please enter a positive number.\n");
20     else {
21         int result = sumdigits(n);
22         printf("Sum of digits of %d is %d\n", n, result);
23     }
24
25     return 0;
26 }
27
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  T
PS C:\Users\aryan> & 'c:\Users\aryan\
in=Microsoft-MIEngine-In-os0t55ib.p1
icrosoft-MIEngine-Pid-hmjkg5di.mrp'
Enter a number: 2468
Sum of digits of 2468 is 20
```